



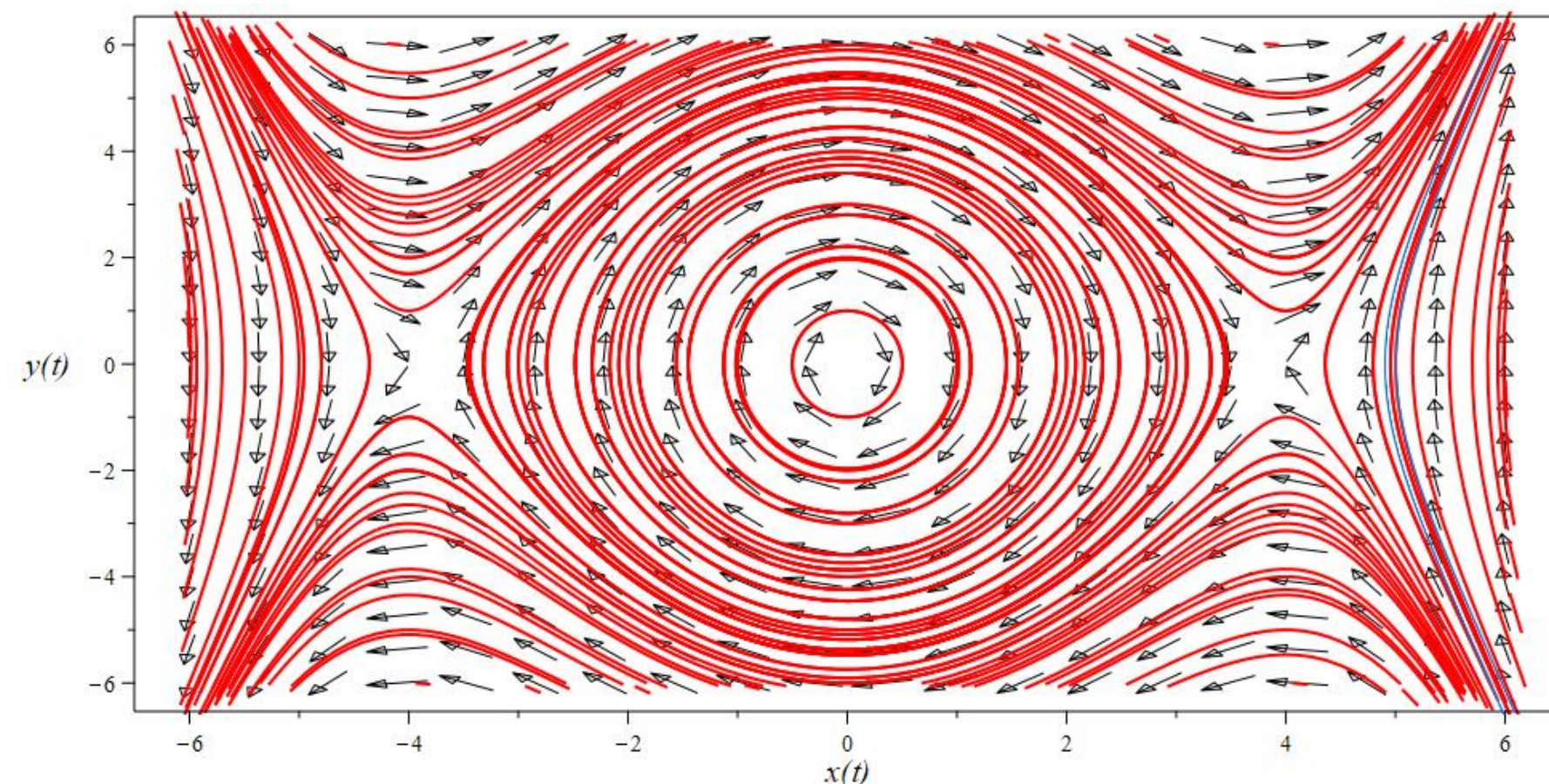
OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Undergraduate Research

at the School of Mathematical and Statistical Sciences, University of Galway

This poster presents a small selection of the images created by undergraduate students in the School of Mathematical and Statistical Sciences, as part of their final year projects. They show the diversity of topics our students are researching, covering numerous areas of pure and applied mathematics, statistics, bioinformatics, and computing.

THOMAS O'CONNOR
The Duffing Equation



DARA KELLY AND CRONAN WARD
Modelling Extremes in Finance and Investigating Asymptotic Dependence

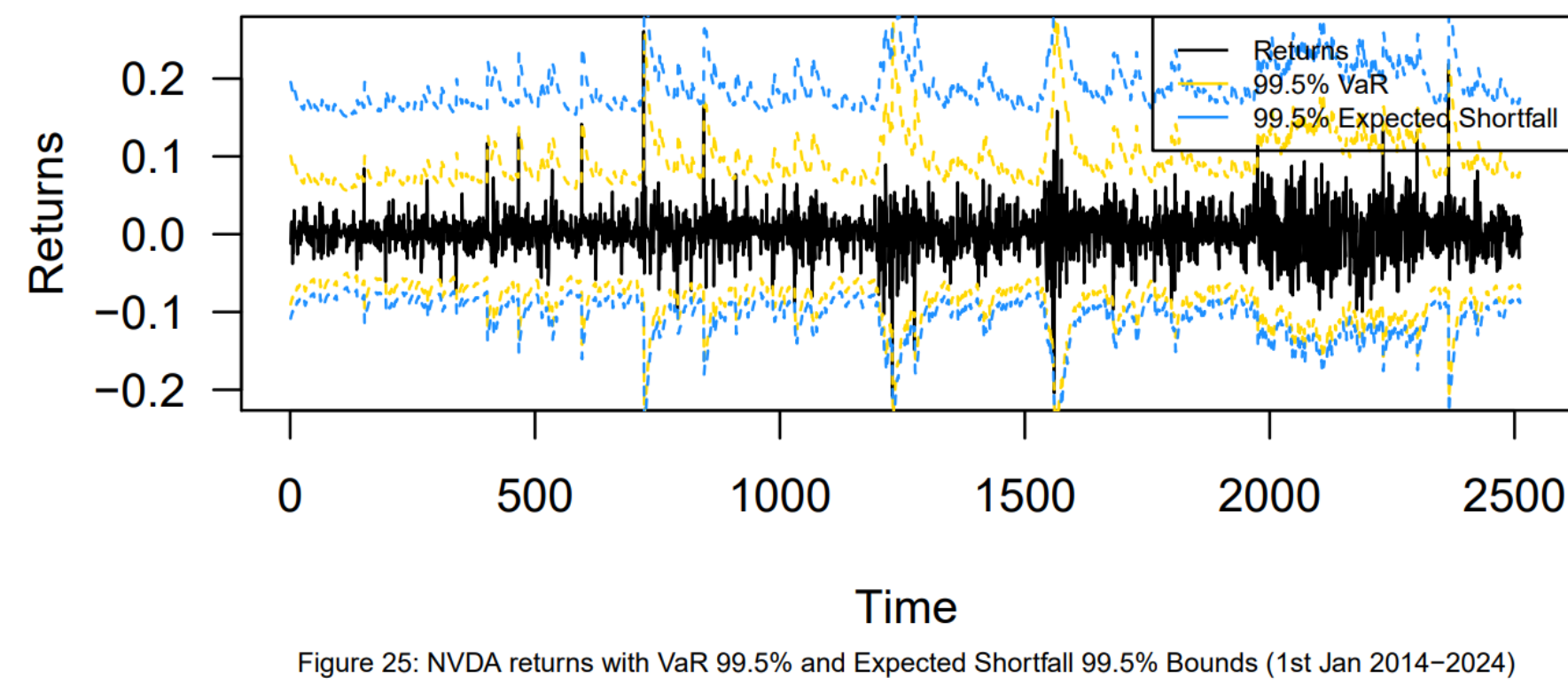
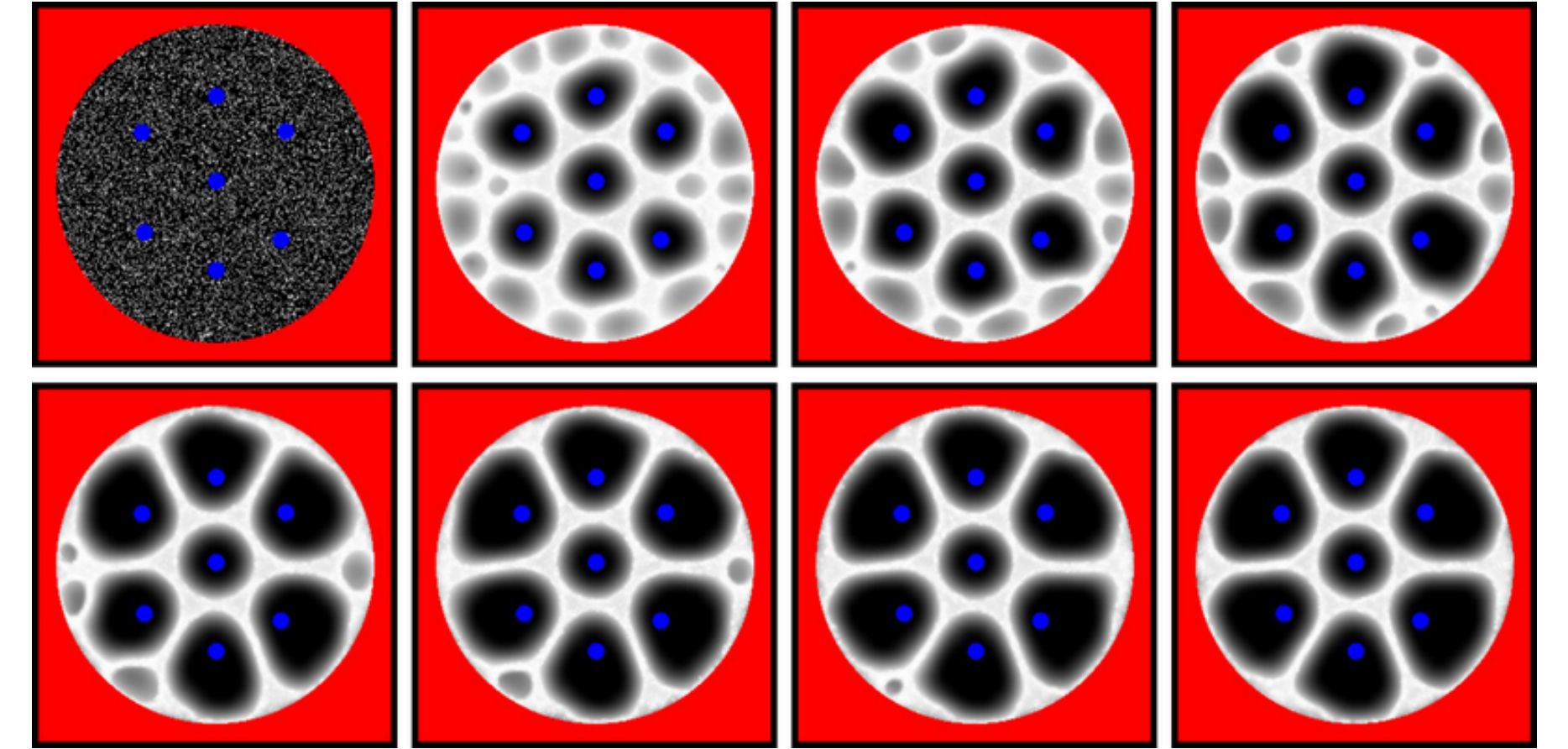
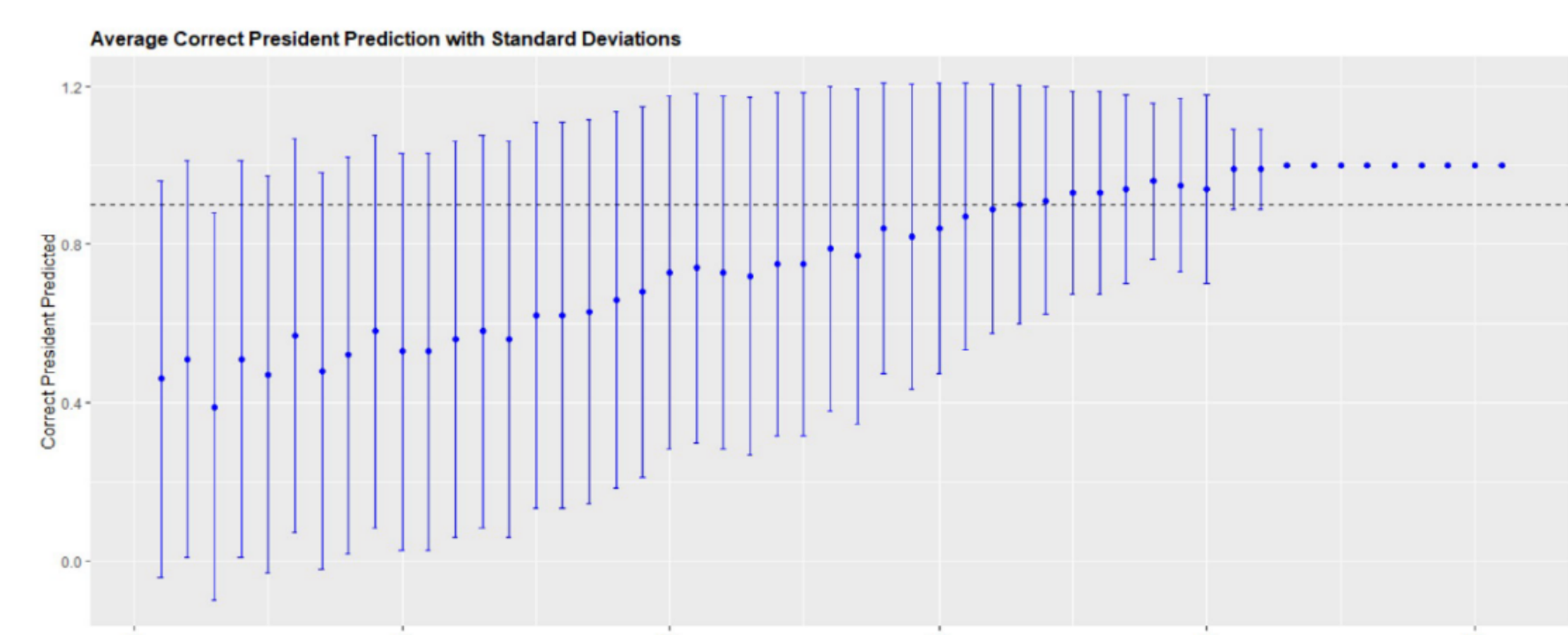


Figure 25: NVDA returns with VaR 99.5% and Expected Shortfall 99.5% Bounds (1st Jan 2014-2024)

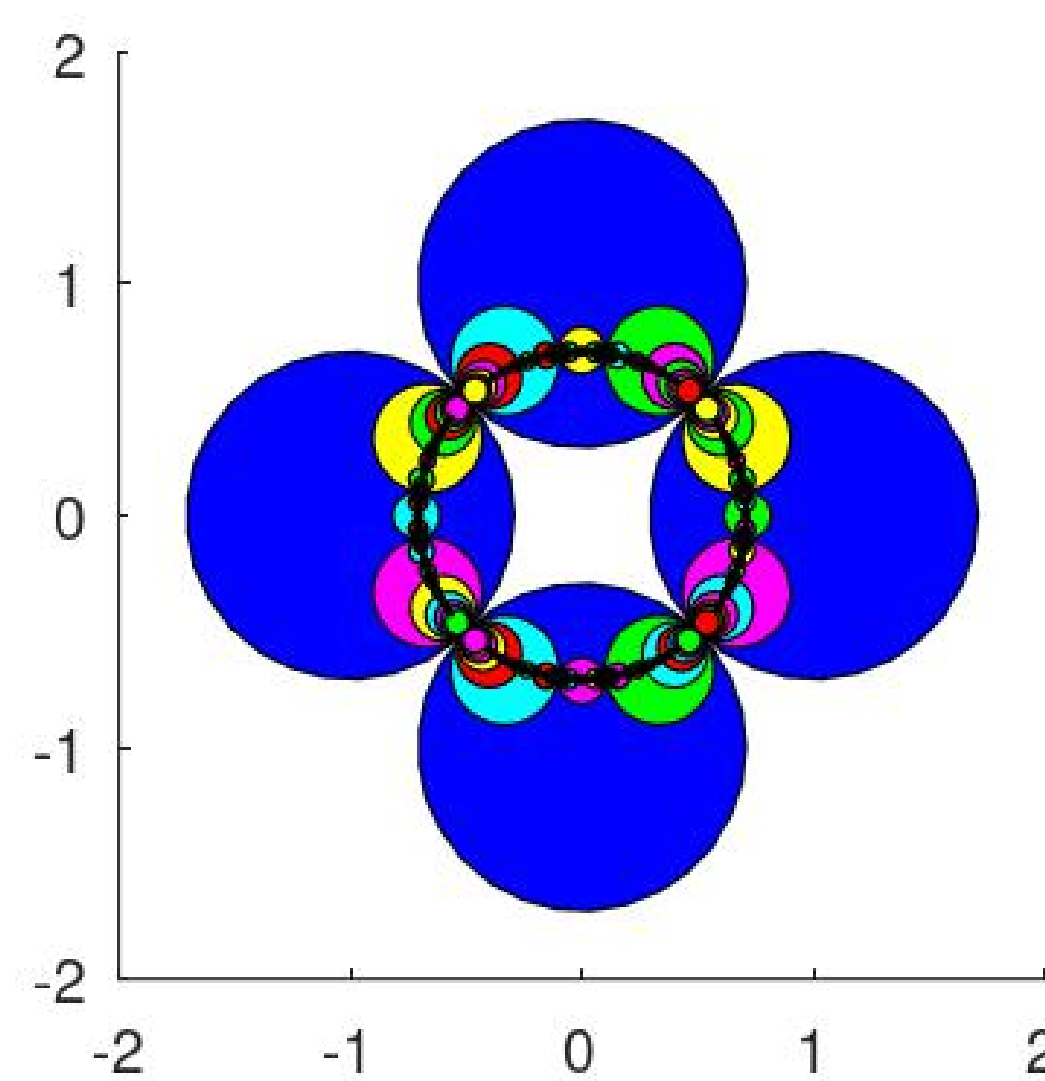
SEAN JANSON AND DMYTRO LYUBKA
Unveiling Emergent Patterns and Euclidean Approximations using Physarum Polycephalum



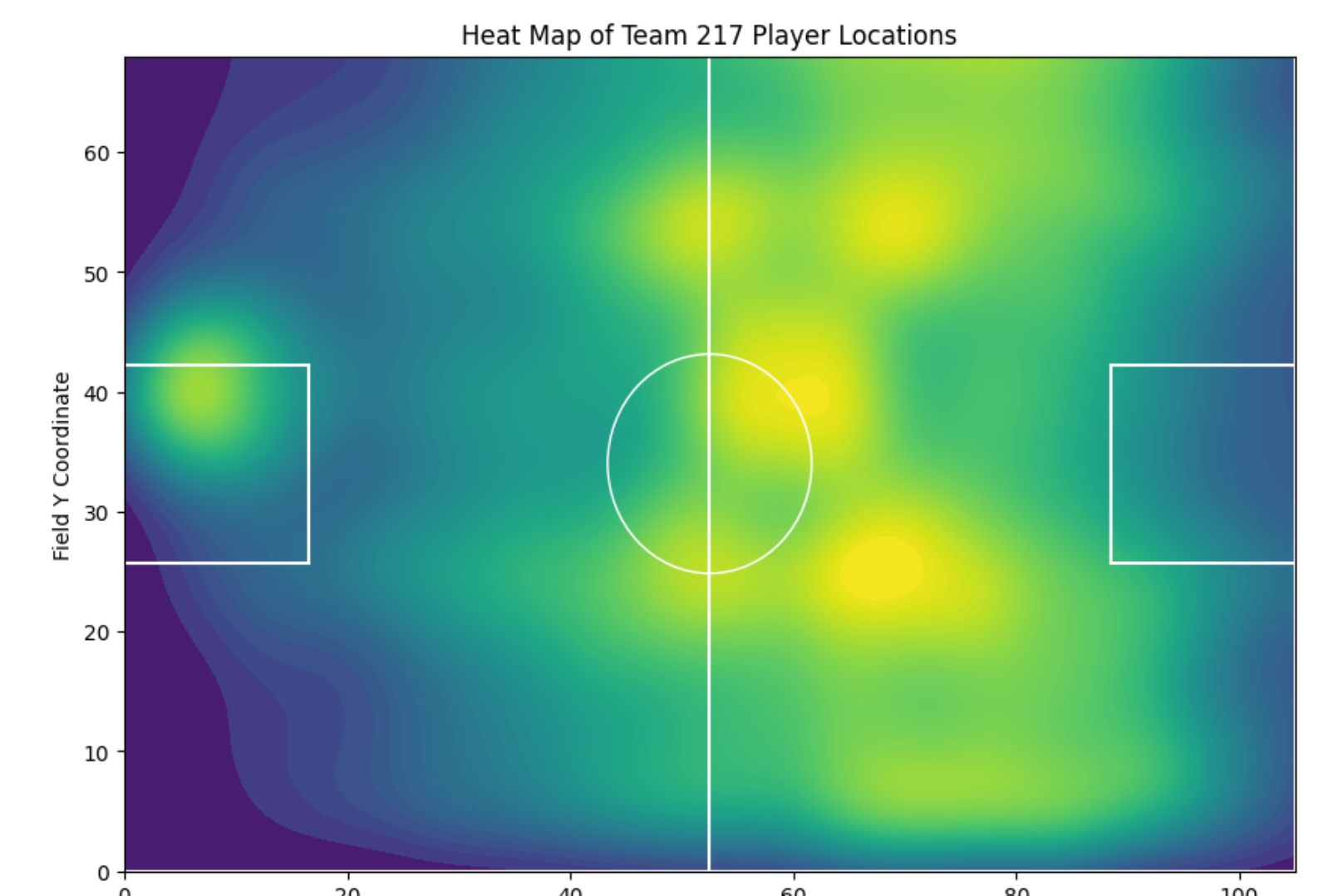
LISE WALL AND EMMA MEANEY
Predicting the US president election using network approaches



ANDREW BELL AND SHANE COTTER
Fractals and Mobius Maps



AILBHE KEANE, KAREN GILLOOLY AND PAULINA KRUNGLEVICIUTE
Analysis of Expected Goals in La Liga 2019/2020



KATELYN MCMAHON AND MATTHEW TALBOT
The Impact of County Demographics and Major Events on Votes for the Democratic Party

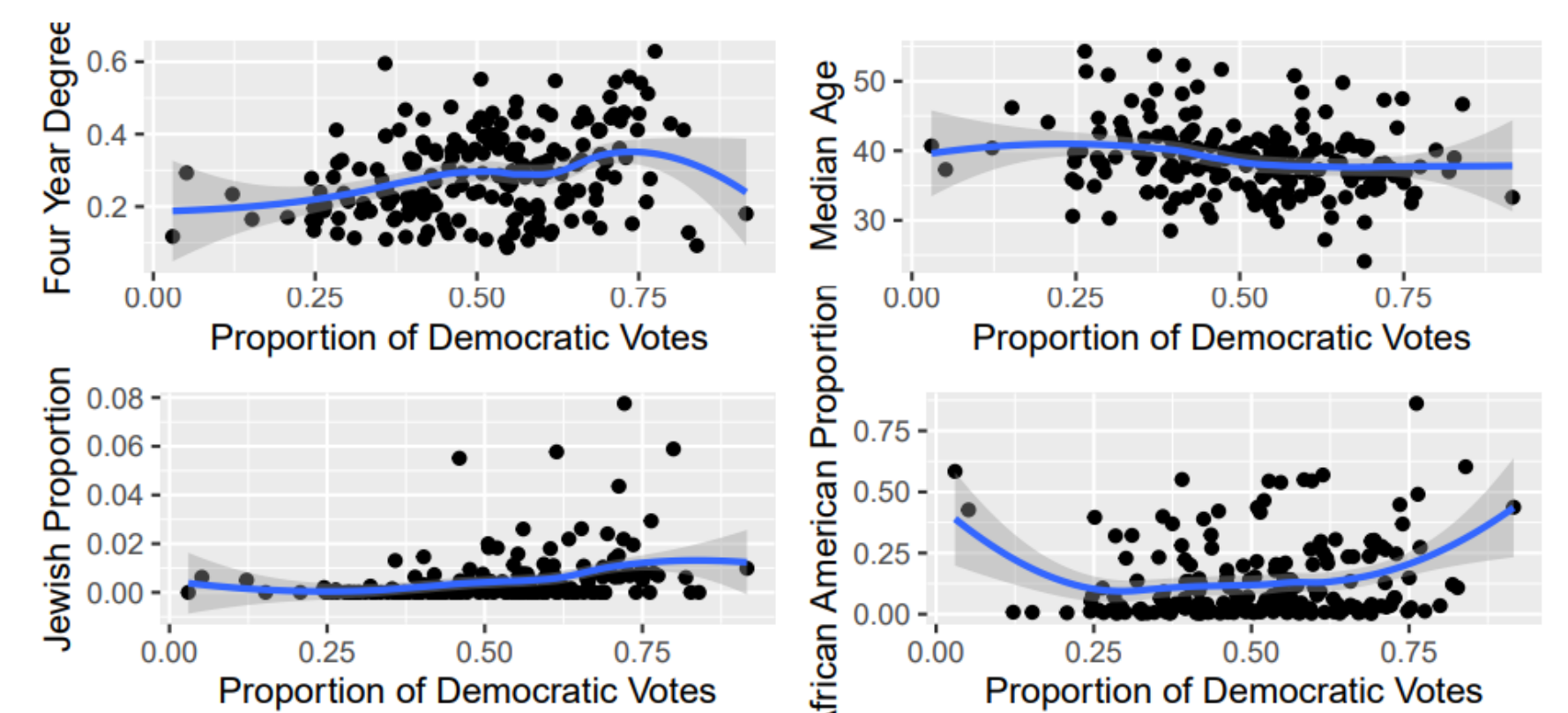
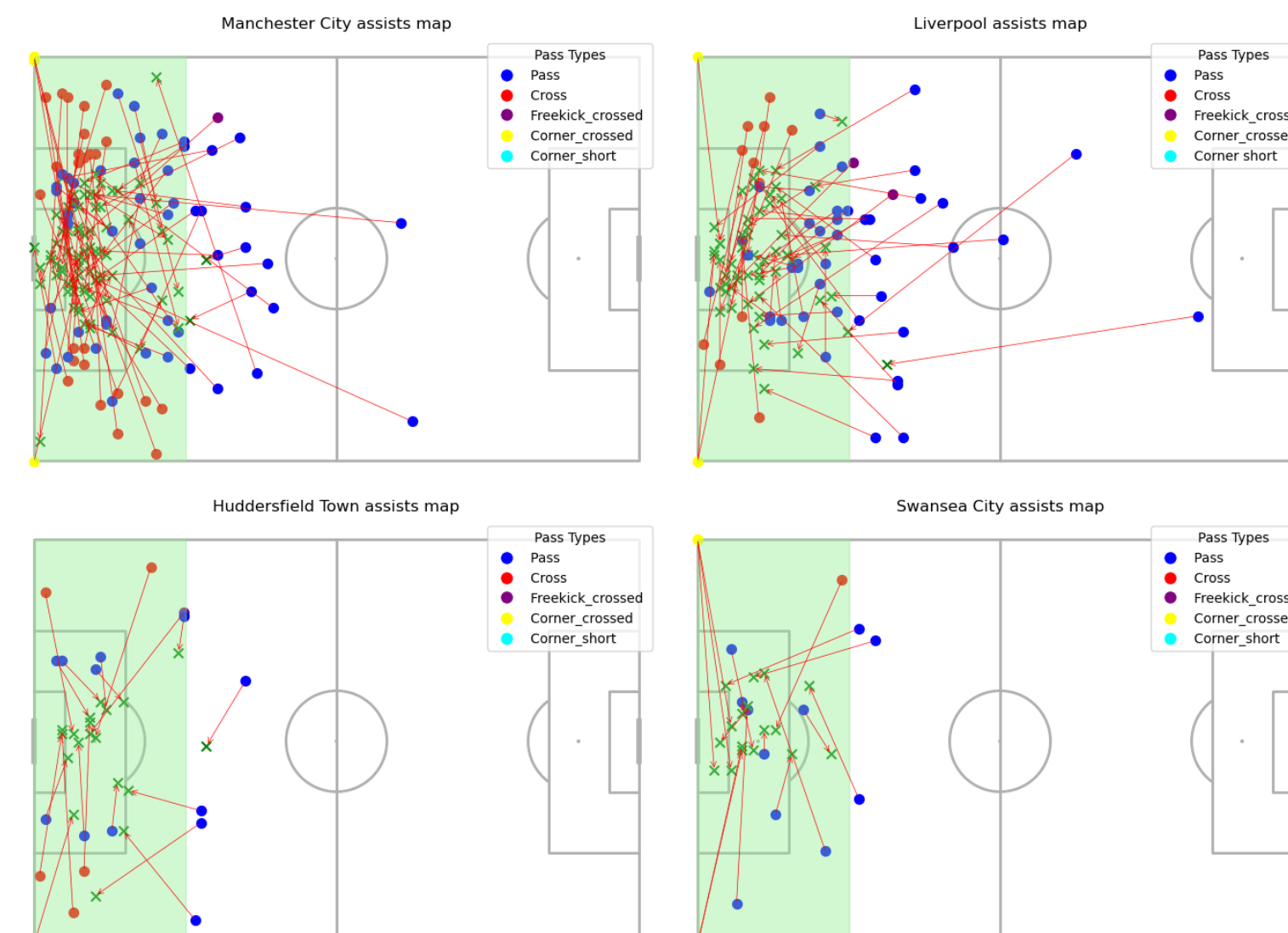
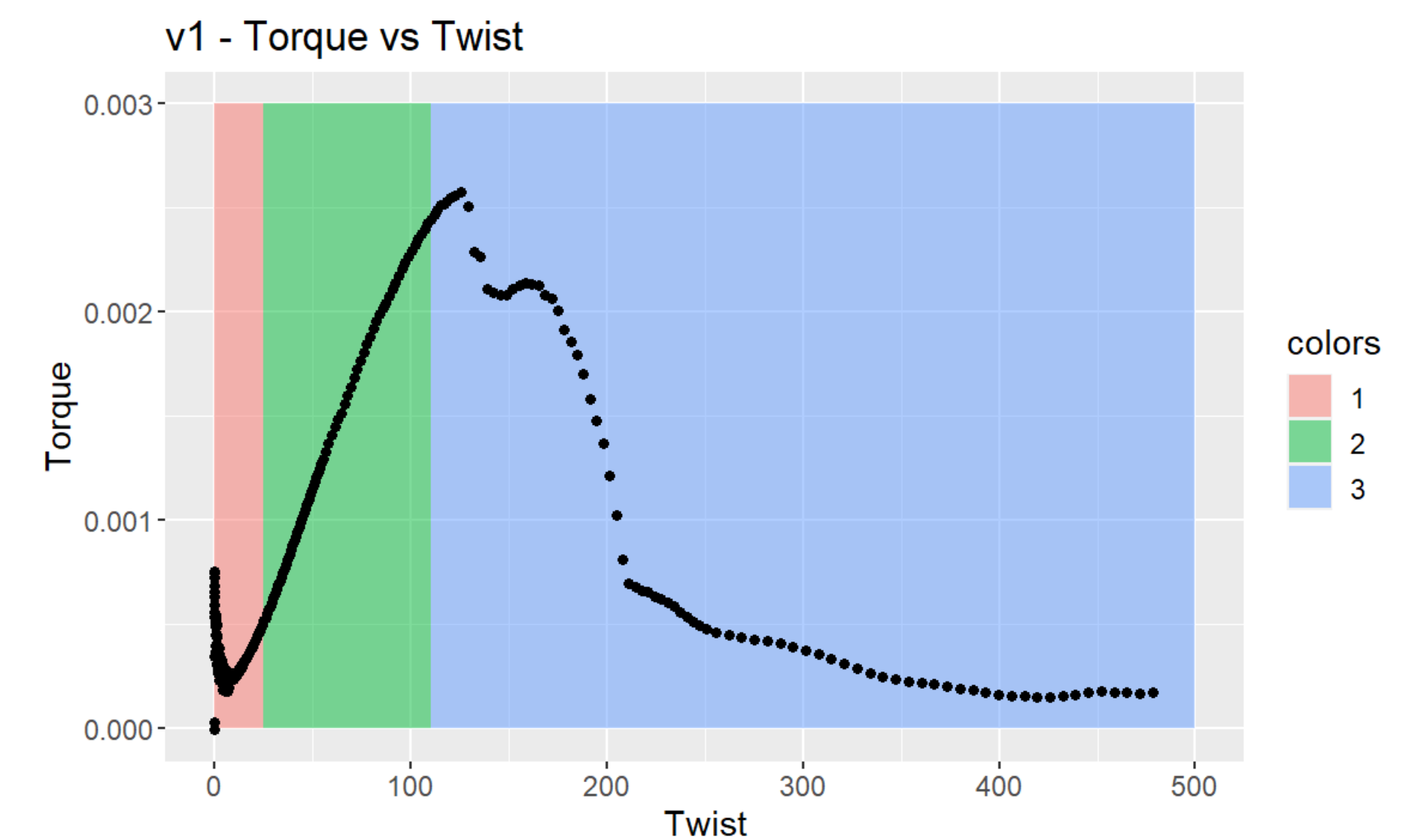


Figure 3: Relationship between the proportion of Democratic votes in a senatorial election and some predictor variables.

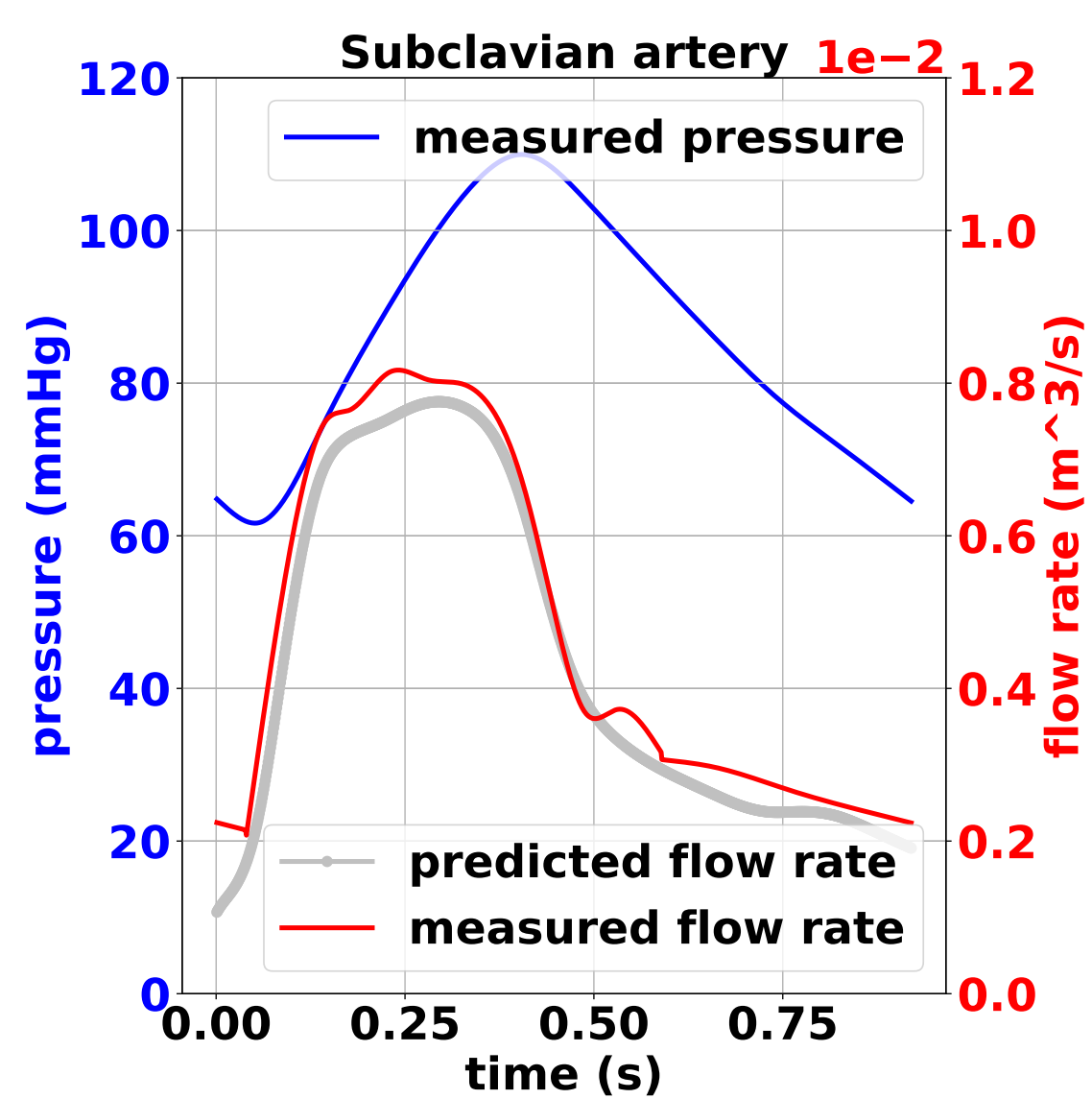
CIAN HUGHES AND CONAL O'MAHONY
Investigating the evolution of tactics in soccer through data analysis



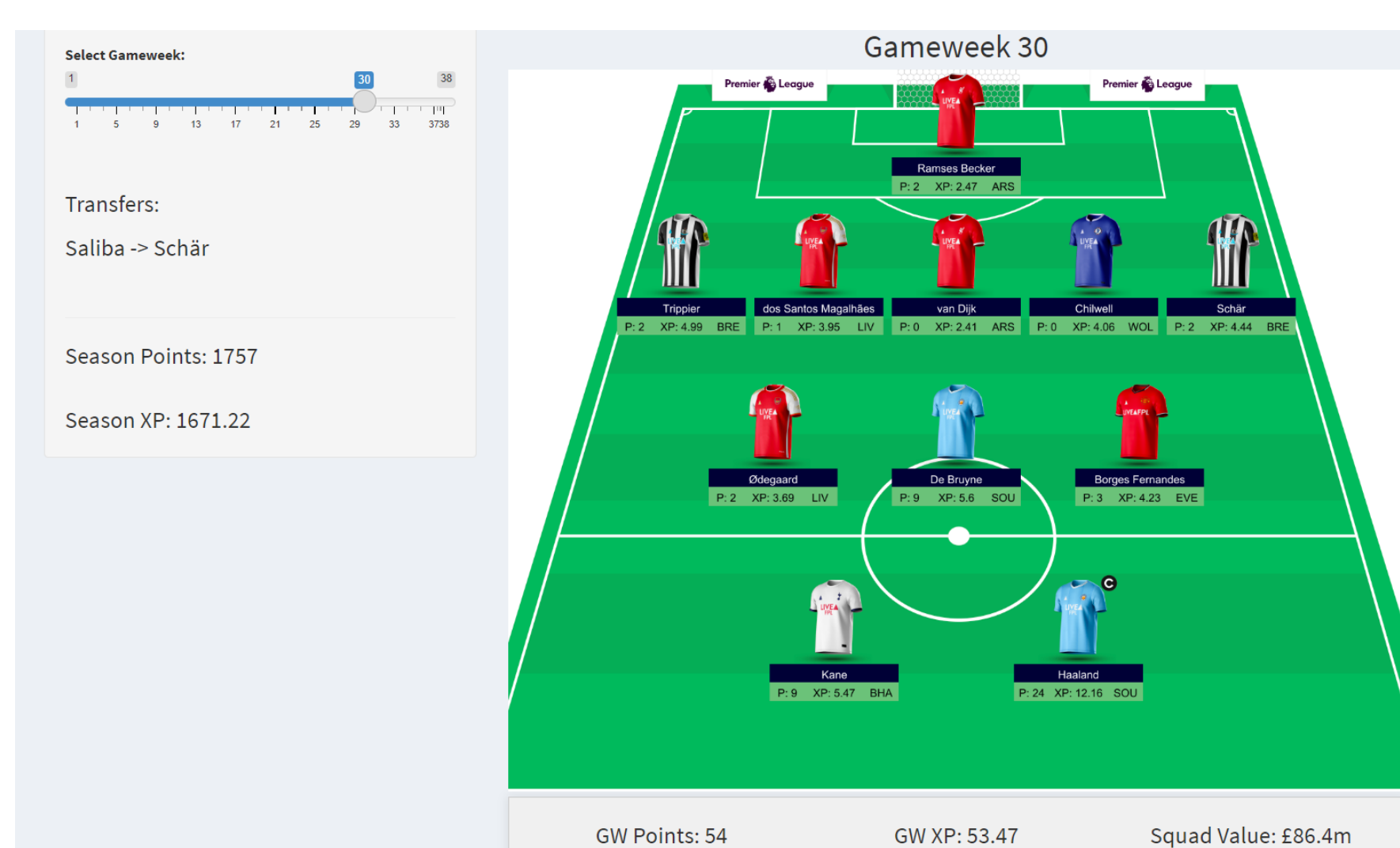
JULIA KOMPANOWSKA AND MAEVE SAMALI
Brain Mechanics



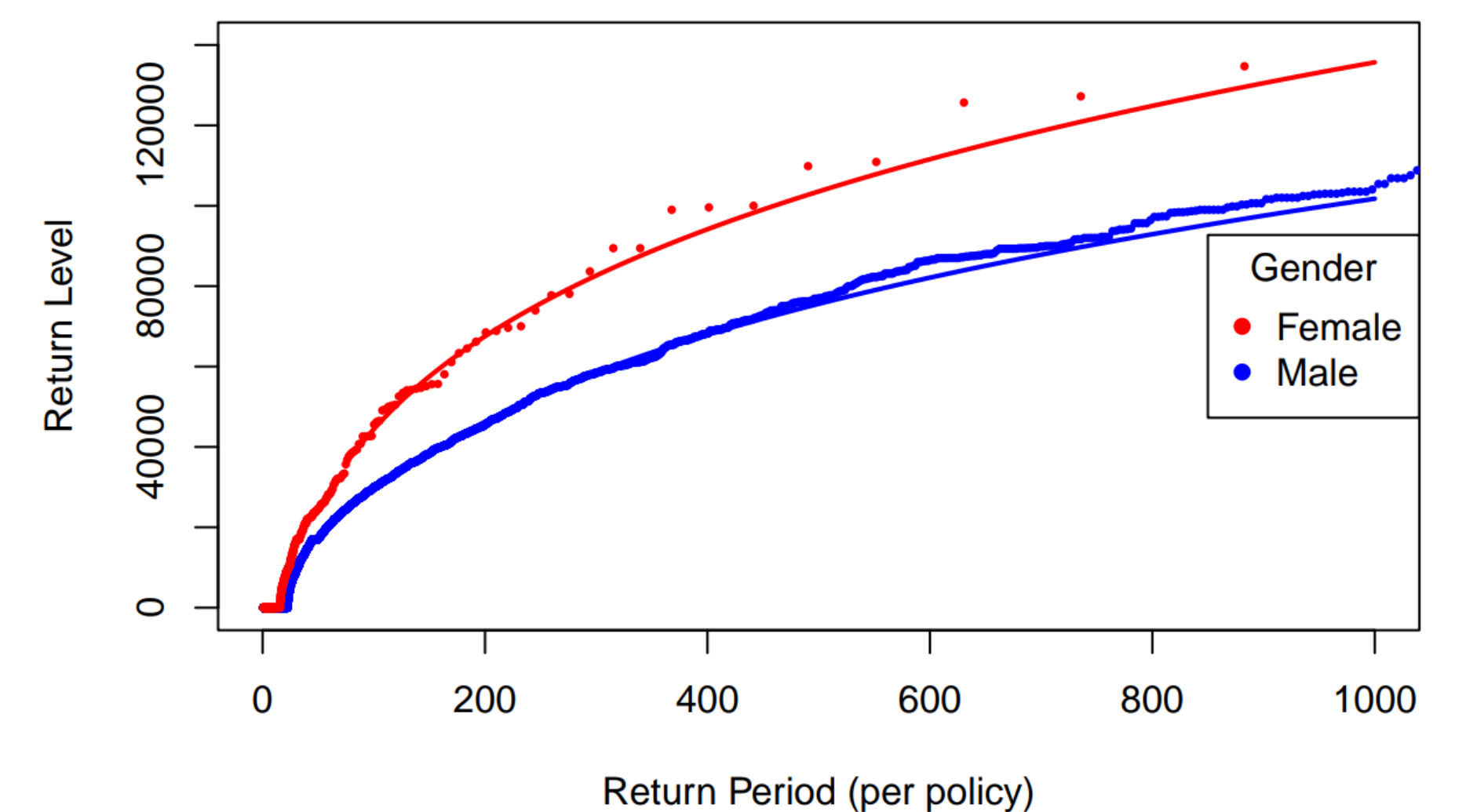
CIARÁN CAMPBELL
The Windkessel Model



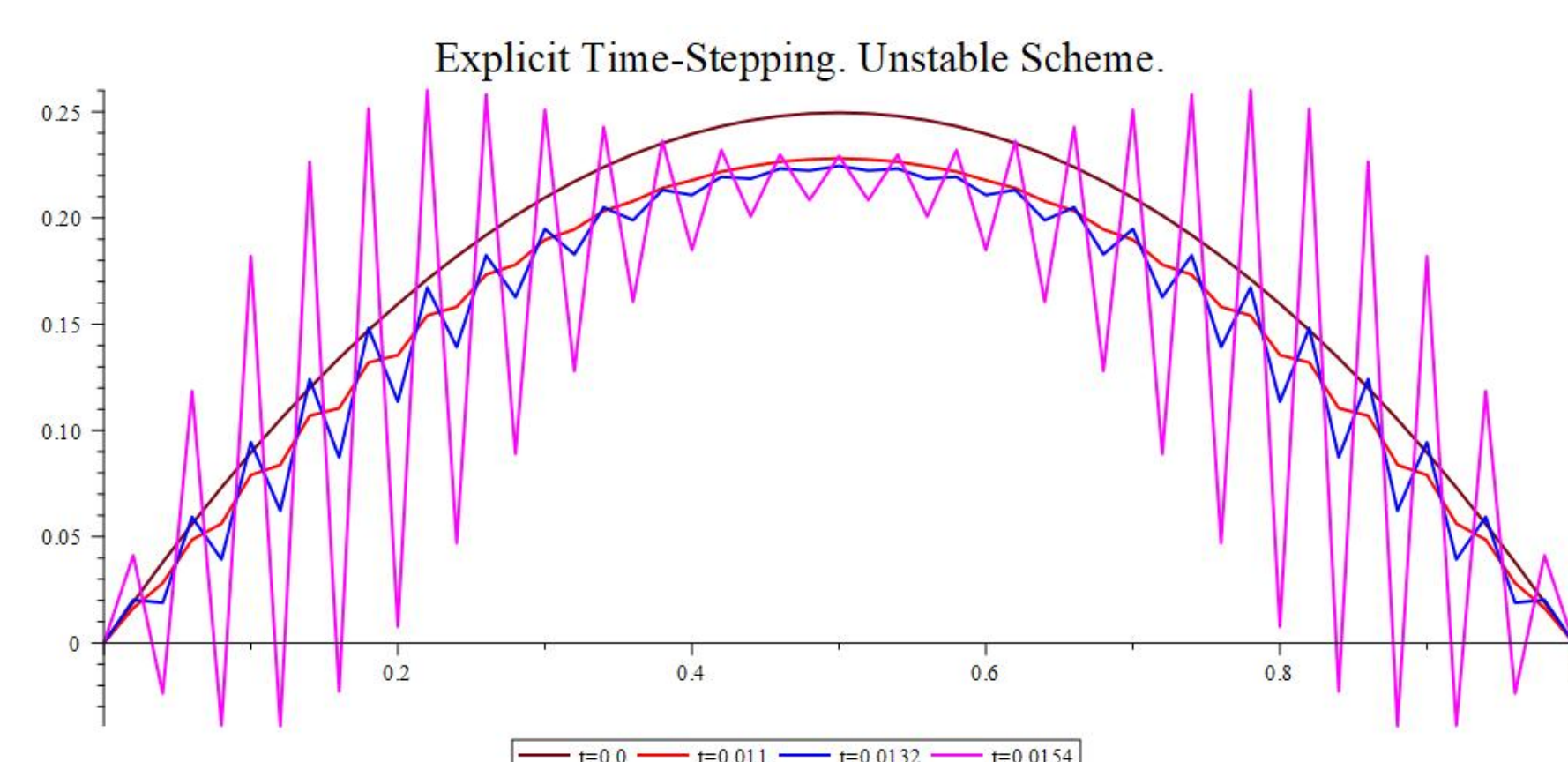
NIAL CAHALAN AND DAIRE ELBERSE
Statistical Modelling of Fantasy Premier League Data



CLAIRE NOLAN AND LEAH WYSE
Analysing Risk Factors in Automobile Insurance Policies



ELI SHEEDY
Finite Difference Methods for PDEs



JAMES NOHILLY
CAOIMHE DOHERTY MCLAUGHLIN
Group Theoretic Analysis of Music

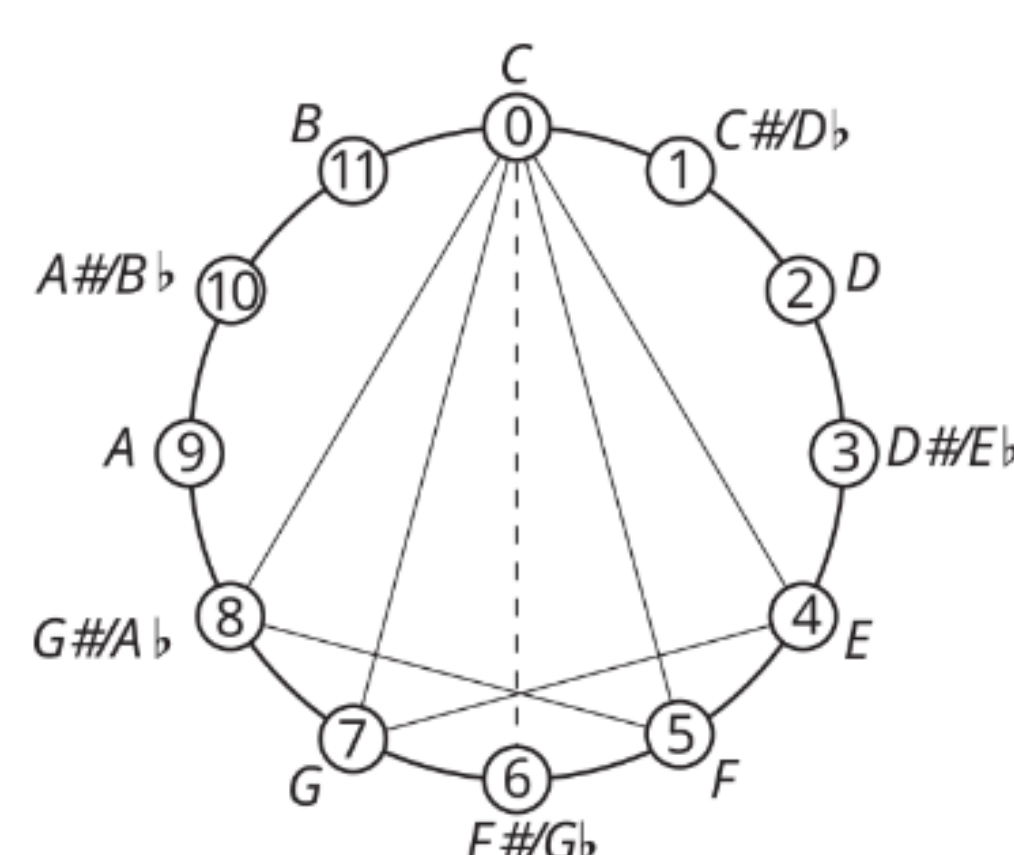


Figure 2.3: I_0 applied to a C-Major triad [4]

$$I_0(0, 4, 7) = (0, 5, 8) = \text{E minor}$$

PAUL CASSIDY
Probability distribution for discrete time alternating lazy quantum walk after 100 steps

